

Def. A map $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{C} : (m, n) \mapsto a_{mn}$

is called a Double Sequence.

observe that: define a set $X \stackrel{\text{def}}{=} \{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{N}\}$.

For a Continuous map $f: X \rightarrow \mathbb{R}$, a sequence $a_n = f(\frac{1}{n})$ converges to $f(0)$. Conversely, for a Convergent sequence $\{a_n\}$, the map f defined on X as

$$f(0) = \lim_{n \rightarrow \infty} a_n, \quad f(\frac{1}{n}) = a_n \quad (n \in \mathbb{N})$$

is Continuous.

Thm. Let $\{a_{mn}\}$ be a double sequence. If $\{a_{mn}\}$ satisfies the followings:

1) For each $m \in \mathbb{N}$, $a_{mn} \rightarrow b_m$ as $n \rightarrow \infty$

2) For each $n \in \mathbb{N}$, $a_{mn} \rightarrow c_n$ as $m \rightarrow \infty$.

3) For any $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $m \geq N \Rightarrow \sup_{n \in \mathbb{N}} |a_{mn} - c_n| \leq \epsilon$.

Then,
$$\lim_{m \rightarrow \infty} b_m = \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} a_{mn} = \lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} a_{mn} = \lim_{n \rightarrow \infty} c_n$$

Proof. For each $m \in \mathbb{N}$, define f_m on $\{\frac{1}{n} \mid n \in \mathbb{N}\}$ as $f_m(\frac{1}{n}) = a_{mn} \quad (n \in \mathbb{N})$

Then, we can observe that

1) $\Leftrightarrow$ For each $m \in \mathbb{N}$, $\lim_{t \rightarrow 0} f_m(t) = b_m$

2) $\Leftrightarrow f_m$ converges pointwise to f , where $f(\frac{1}{n}) = c_n \quad (n \in \mathbb{N})$

3) $\Leftrightarrow f_m$ converges uniformly to f .

Therefore,
$$\lim_{m \rightarrow \infty} b_m = \lim_{m \rightarrow \infty} \lim_{t \rightarrow 0} f_m(t) = \lim_{t \rightarrow 0} \lim_{m \rightarrow \infty} f_m(t) = \lim_{t \rightarrow 0} f(t) = \lim_{n \rightarrow \infty} f(\frac{1}{n}) = \lim_{n \rightarrow \infty} c_n.$$

Lemma. Let $f_n: X \rightarrow Y$ be continuous functions s.t. $f_n \rightarrow f$.

If a sequence $\{x_n\}$ in X converges to $x \in X$, then $\{f_n(x_n)\}$ converges to $f(x)$.

Proof. Since $f_n \rightarrow f$ and f_n is continuous, f is continuous.

Given $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $n \geq N_1 \Rightarrow \sup_x |f_n - f| < \frac{\epsilon}{2}$.

Meanwhile, for these $n \geq N_1$, there is $N_2 \in \mathbb{N}$ s.t. $m \geq N_2 \Rightarrow |f_n(x_m) - f_n(x)| < \frac{\epsilon}{2}$.

Now, for $k \geq \max\{N_1, N_2\}$,

$$|f_k(x_k) - f(x)| \leq |f_k(x_k) - f_k(x)| + |f_k(x) - f(x)| < \epsilon. \quad \square$$

Thm. Let $\{a_{mn}\}$ be a double sequence s.t.

1) For each $m=1, 2, \dots$, $\sum_{n=1}^{\infty} |a_{mn}| = b_m$

2) $\sum_{m=1}^{\infty} b_m$ converges.

Then,
$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{mn} = \lim_{N \rightarrow \infty} \sum_{m,n=1}^N a_{mn}. \quad \dots (*)$$

Proof. Let $X = \{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{N}\}$, and for each $m \in \mathbb{N}$, define

$$f_m: X \rightarrow \mathbb{R}; x \mapsto \begin{cases} \sum_{j=1}^{\infty} a_{mj} & \text{if } x=0 \\ \sum_{j=1}^n a_{mj} & \text{if } x=\frac{1}{n}, n \in \mathbb{N}. \end{cases} \quad \left(\text{That is, } f_m\left(\frac{1}{n}\right) = \sum_{j=1}^n a_{mj} \right)$$

Weierstrass M-test $\downarrow$

and let $g_n = \sum_{m=1}^n f_m$. Since $|f_m| \leq \sum_{j=1}^{\infty} |a_{mj}| = b_m$ and $\sum b_m$ converges, g_n converges uniformly.

And, each f_m is continuous, thus $g_n = \sum_{m=1}^n f_m$ is continuous, thus $g = \lim_{n \rightarrow \infty} g_n$ is continuous, therefore

$$\lim_{n \rightarrow \infty} g\left(\frac{1}{n}\right) = g(0) = \lim_{n \rightarrow \infty} g_n\left(\frac{1}{n}\right).$$

$\downarrow$ g -Conti. $\downarrow$ $n \rightarrow \infty$ $\downarrow$ Lemma.

$$\lim_{n \rightarrow \infty} g\left(\frac{1}{n}\right) = \lim_{n \rightarrow \infty} \sum_{m=1}^{\infty} f_m\left(\frac{1}{n}\right) = \lim_{n \rightarrow \infty} \sum_{m=1}^{\infty} \sum_{j=1}^n a_{mj} = \lim_{n \rightarrow \infty} \sum_{m=1}^{\infty} \sum_{j=1}^{\infty} a_{mj} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{mn}$$

$$g(0) = \sum_{m=1}^{\infty} f_m(0) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} \quad \text{In finite sum, convergences preserved limit.}$$

$$\lim_{N \rightarrow \infty} g_N\left(\frac{1}{N}\right) = \lim_{N \rightarrow \infty} \sum_{m=1}^N f_m\left(\frac{1}{N}\right) = \lim_{N \rightarrow \infty} \sum_{m=1}^N \sum_{j=1}^N a_{mj} = \lim_{N \rightarrow \infty} \sum_{m,n=1}^N a_{mn}.$$

Note: If $(*)$ holds, then we say that sum of $\{a_{mn}\}$ can be exchanged.

Thm. If $\{a_{mn}\}$ is positive, then sum of $\{a_{mn}\}$ can be exchanged.

Proof. For any $M, N \in \mathbb{N}$ s.t. $M \leq N$, since $a_{mn} \geq 0$,

$$\sum_{m=1}^M \sum_{n=1}^N a_{mn} \leq \sum_{m,n=1}^N a_{mn}$$

Taking limit to the both side, then

$$\sum_{m=1}^M \sum_{n=1}^{\infty} a_{mn} = \lim_{N \rightarrow \infty} \sum_{m=1}^M \sum_{n=1}^N a_{mn} \leq \lim_{N \rightarrow \infty} \sum_{m,n=1}^N a_{mn}$$

Since M was arbitrary,

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} \leq \sum_{m,n=1}^{\infty} a_{mn}.$$

And, the converse inequality is given by: for any $N \in \mathbb{N}$

$$\sum_{m,n=1}^N a_{mn} \leq \sum_{n=1}^N \sum_{m=1}^{\infty} a_{mn}.$$

Corollary. If $a_k \geq 0$ for all $k \in \mathbb{Z}$, then

$$\lim_{n \rightarrow \infty} \sum_{k=-n}^n a_k = \lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} \sum_{k=-m}^n a_k = \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \sum_{k=-m}^n a_k.$$